Axle Informatics

Business Requirements Document

Virtual Laboratory Freezer Management System

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1. Introduction

1.1 Document Purpose

This Business Requirements Document defines the business problem, objectives, scope, stakeholder needs, business processes, business and functional requirements, business rules, information needs, risks, and acceptance expectations for the Virtual Laboratory Freezer Management System. It is intended to establish a common business baseline before detailed architecture, design, implementation, and testing are produced.

1.2 Project Background

Laboratory staff currently spend significant time physically visiting laboratory spaces and checking freezers to determine where plates, samples, specimens, and other materials can be stored. Storage is distributed across multiple physical locations, buildings, floors, rooms, freezers, shelves, racks, and boxes. The current process does not provide a centralized virtual view of storage availability, making it difficult to determine suitable space before traveling to the lab.

1.3 Problem Statement

Laboratory staff need a centralized way to view freezer storage and identify available capacity without physically inspecting each freezer. Administrators need a centralized platform to manage freezers across locations. The solution must make occupied and available storage visible, allow space to be reserved before physical storage, reduce manual searching and storage conflicts, and make laboratory storage searchable and accessible from anywhere permitted by organizational policy.

1.4 Proposed Business Solution

The proposed solution is a centralized, agent led virtual laboratory storage management platform. Staff will describe their storage needs to a conversational agent, which can ask clarifying questions and use authoritative backend data to identify compatible available space and recommend the physically closest suitable freezer. The agent will present a proposed reservation and obtain the user's explicit confirmation before the backend reserves the space. Users will also be able to ask the agent for an item's current freezer and complete storage path. The platform will maintain the physical storage hierarchy and storage status, support scan-assisted check-in and check-out, handle assistance requests, and allow authorized administrators to manage freezer configurations and restrictions through the agent.

2. Business Need, Objectives, and Benefits

2.1 Business Need

- Provide centralized visibility into laboratory freezer storage across multiple physical locations.
- Reduce staff time spent walking to laboratories solely to inspect available storage.
- Provide a reliable method to identify storage that is physically large enough and compatible with material requirements.
- Prevent storage conflicts caused by multiple users attempting to use the same space.
- Keep the virtual representation of storage aligned with physical check-in and check-out activity.
- Allow administrators to manage heterogeneous freezer configurations from one system.

2.2 Business Objectives

- Enable staff to determine suitable available freezer space remotely before visiting the laboratory.
- Ensure storage recommendations account for capacity, temperature, storage restrictions, and access restrictions.
- Recommend the physically closest suitable available freezer, including options on another floor, in another building, or at another location when necessary.
- Require reservation before physical storage and automatically release unused reservations after two days.
- Prevent duplicate reservations and handle simultaneous reservation attempts without creating inconsistent storage state.
- Support scan-assisted item placement and removal so storage status can be updated as physical actions occur.
- Provide a conversational interface through which staff can describe storage requirements and receive suitable options.
- Support future programmatic requests from other systems without committing this BRD to a specific integration technology.

2.3 Expected Business Benefits

- Fewer unnecessary trips to freezer locations.
- Less manual searching and reduced staff time spent locating storage.
- Lower risk of double booking or placing material in unsuitable storage.
- Improved visibility and utilization of freezer capacity.
- Improved traceability of who reserved, checked in, or checked out stored material.
- More consistent operational handling of storage across multiple locations and freezer configurations.

2.4 Initial Success Measures

The following are qualitative success measures for the first BRD draft. Quantitative targets should be agreed with the stakeholder during validation:

- Staff can identify suitable storage without first performing a physical freezer inspection.
- The system does not permit two active reservations for the same reservable storage position.
- The virtual storage state is updated through the defined check-in/check-out workflow.
- Administrators can centrally manage freezer structures and restricted storage.
- Users can provide storage requirements conversationally and receive recommendations derived from current backend storage data.

3. Project Scope

3.1 In Scope

- Centralized representation of storage across Location → Building → Floor → Room → Freezer → Shelf → Rack → Box, with individual positions/slots represented where applicable.
- Support for freezers with different structures, capacities, temperature ranges, and storage restrictions.
- Support for storage areas intended for larger objects and determination of whether sufficient capacity exists.
- Storage discovery based on material requirements and available capacity.
- Recommendation of the physically closest suitable available freezer by the agent.
- Reservation of storage before physical placement.
- Automatic cancellation of an unused reservation after two days and release of the space.
- Conflict prevention when multiple users attempt to reserve the same storage.
- Scan-assisted check-in and check-out during physical storage operations.
- Relocation through checkout from the existing location followed by reservation and check-in at a new location.
- Role-based access for Staff, Project Lead, and Administrator users.
- Administrator management of freezer structures and restricted/sensitive storage.
- Assistance requests when a user needs another person to perform storage, retrieval, or transfer activity handled by the agent.
- Conversational agent for capturing storage needs and retrieving suitable options from the system.
- Capability for the agent/service layer to support future requests originating from other systems.

3.2 Out of Scope / Not Yet Confirmed

The stakeholder interview did not explicitly define a complete list of project exclusions. Therefore, this draft does not invent functional exclusions. The following implementation decisions are intentionally outside the business-level scope of this BRD and will be decided in later SDLC phases:

- Specific cloud provider, programming language, database, frontend framework, or deployment architecture.
- Specific conversational agent vendor or model.
- Use of Claude or Model Context Protocol (MCP) as an implementation mechanism; these were mentioned as possible technical directions, not mandated business requirements.
- Detailed API contracts, database schema, sequence diagrams, infrastructure design, and test implementation.
- The authorized personnel to do the storage or item transfer on request, has not been included as a stakeholder of the system.

3.3 Business Boundary

The system manages the digital representation and workflow of laboratory storage. Physical placement, retrieval, and movement of laboratory material remain human activities. The system supports and records these activities through reservations, scanning, check-in/check-out, and assistance requests; it does not physically move materials.

4. Stakeholders and Business Users

4.1 Stakeholder Overview

Stakeholder / Role	Primary Business Need	Key Responsibilities / Interest	Validation Status
Laboratory Staff	Quickly find and use appropriate storage.	Provide storage requirements; find/reserve space; check items in/out; scan during physical operations; raise assistance requests.	Confirmed
Project Lead	Oversee the development and operational effectiveness of the solution.	Review system behavior and usage; participate in requirement/change decisions; broader operational visibility. Specific privileged actions remain to be confirmed.	Partly proposed
Administrator	Centrally manage freezer infrastructure and sensitive storage.	Add/remove/modify freezer structures and configurations; manage restrictions; maintain storage attributes.	Confirmed
External Systems	Submit programmatic requests to the storage capability/agent.	Integration behavior and supported systems to be defined later.	Confirmed need; details TBD

4.2 User Roles and Access Summary

Capability	Staff	Project Lead	Administrator
View storage hierarchy and availability	Yes	Yes	Yes
Search / request suitable storage	Yes	Yes	Yes
Reserve storage	Yes	Yes	Yes
Check items in/out	Yes	Yes	Yes
Raise assistance request	Yes	Yes	Yes
View broader operational activity	Limited to permitted data	Yes	Yes
Add/remove/modify freezer structures	No	No	Yes
Manage restricted/sensitive freezer configuration	No	No	Yes

5. Current and Future Business Process

5.1 Current-State (As-Is) Process for storage

Staff visit a laboratory and inspect freezers to identify possible space. This creates avoidable trips and relies on local knowledge of current occupancy, physical size, and storage restrictions.

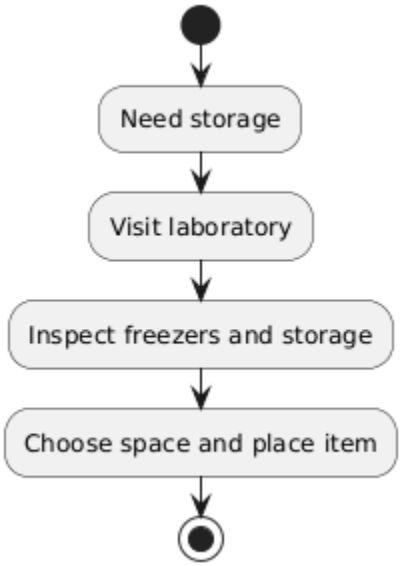


Figure 1 current workflow to store an item

5.2 Future State (To-Be) Process for storage

Staff provide material and storage needs remotely. The system filters incompatible or unavailable space, recommends the physically closest suitable option, reserves the selected space, and records scan-assisted physical placement.

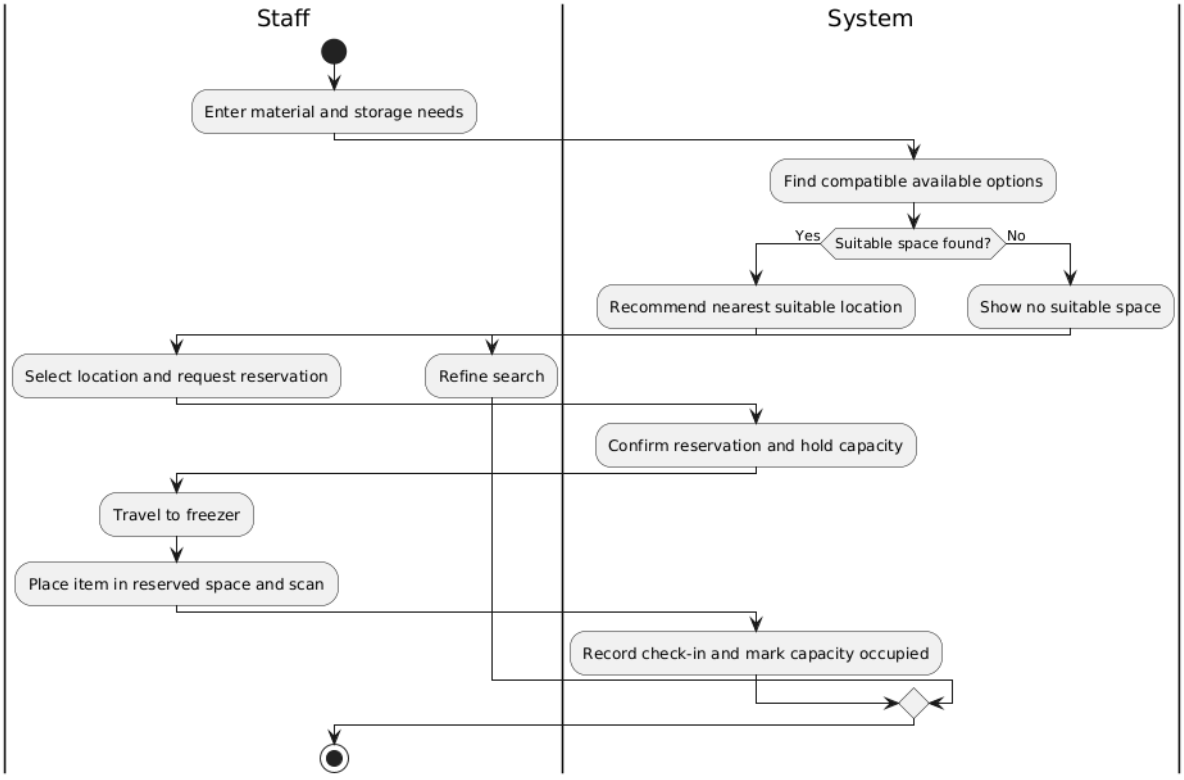


Figure 2 workflow of storing an item with the new system

5.3 Core Business Workflows

5.3.1 Remove an item

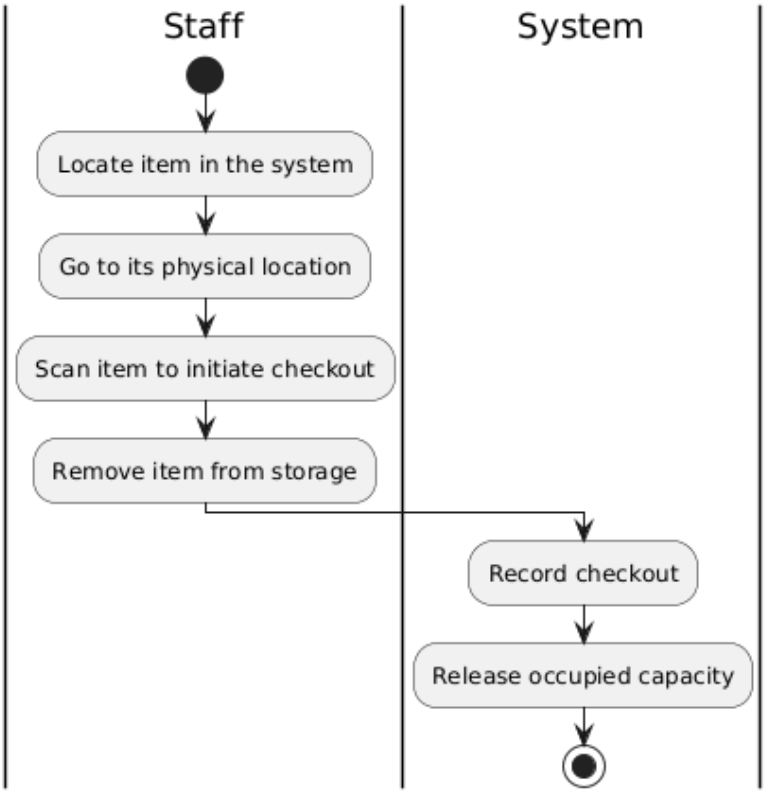


Figure 3 Workflow for removing an item from the freezer

5.3.2 Reservation expiration

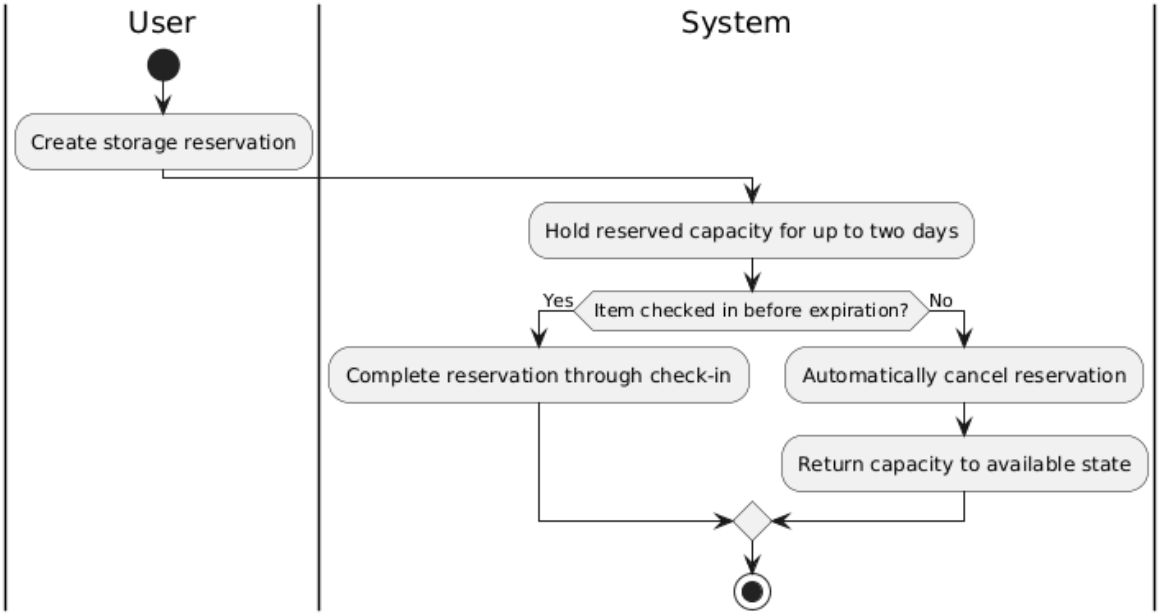


Figure 4 Workflow in case reservation expires

5.3.3 Assistance request

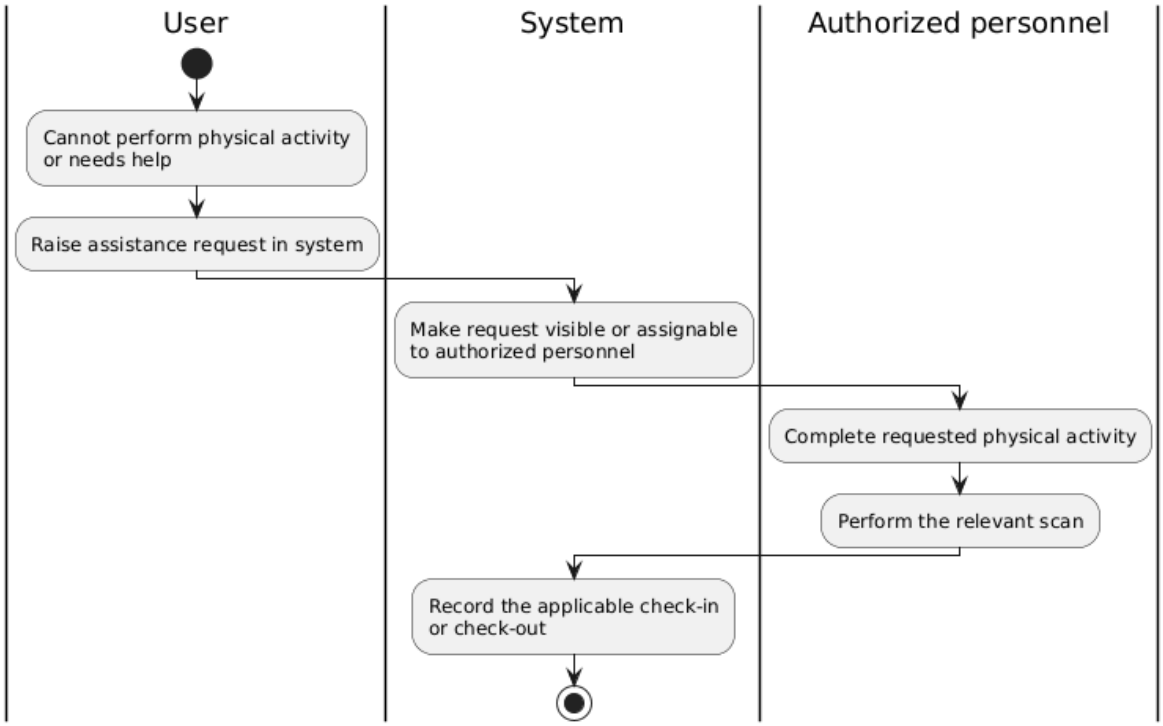


Figure 5 Workflow for requesting assistance to store or move an item

6. Epic Catalog and Requirements

Business requirements state the outcomes expected of the complete system. The epic reference groups related work; it is not a claim that the epic replaces the stakeholder problem statement and interview as the original source. Each requirement retains its earlier identifier where a direct successor exists.

6.1 Epics

Epic	Name	Business outcome
EP-01	Storage Visibility and Structure	Represent variable freezer layouts and make storage state and location available through the agent.
EP-02	Compatible Storage Discovery	Interpret material needs, identify suitable capacity, and recommend the physically closest suitable freezer.
EP-03	Reservation Management	Obtain the user's confirmation, allocate suitable space safely, and release unused holds after two days.
EP-04	Material Check-In Check-Out and Relocation	Keep physical scans, occupancy, item location, and movement history aligned.
EP-05	Assistance Requests	Capture and route requests for authorized personnel to perform physical work.
EP-06	Conversational Storage Assistant	Carry out discovery, clarification, reservation, and stored-item lookup through a conversation grounded in backend data.
EP-07	External System Requests	Allow future authorized programmatic requests through the same backend rules.
EP-08	Administration and Restricted Storage	Manage freezer configurations and enforce access to sensitive storage.

6.2 Business Requirements

ID	Epic reference	Business requirement
BR-01	EP-01, EP-06	The user shall be able to ask the agent about freezer storage across supported locations without visiting the laboratory.
BR-02	EP-01	The system shall be able to represent Location, Building, Floor, Room, and Freezer, followed by the internal storage levels applicable to each freezer, including shelves, racks, baskets, boxes, positions, or larger storage areas where configured.
BR-03	EP-01, EP-06	The user shall be able to obtain available, reserved, occupied, and accessible storage information by asking the agent.
BR-04	EP-02	The user shall be able to describe storage needs for materials of different sizes, including large objects, in conversation with the agent.
BR-05	EP-02, EP-06	The system shall be able to clarify incomplete or ambiguous storage requirements with the user before recommending capacity.
BR-06	EP-02	The user shall be able to receive available storage options that meet size, temperature, material type, and access requirements.

ID	Epic reference	Business requirement
BR-07	EP-02	The user shall be able to receive the physically closest suitable available freezer as the leading recommendation.
BR-08	EP-03, EP-06	The user shall be able to confirm a proposed reservation with the agent before the system reserves space for the item.
BR-09	EP-03	The system shall be able to prevent conflicting active reservations, including during simultaneous requests.
BR-10	EP-03	The system shall be able to cancel an unused reservation after two days and make its capacity available again.
BR-11	EP-04	The user shall be able to check an item in or out through scan-assisted physical operations.
BR-12	EP-01, EP-04, EP-06	The user shall be able to ask the agent where an item is stored and receive its current freezer, full storage path, and relevant placement details.
BR-13	EP-04, EP-06	The user shall be able to relocate an item through check-out, agent-assisted selection and reservation of a destination, and scan-assisted check-in.
BR-14	EP-05, EP-06	The user shall be able to ask the agent for help with physical storage, retrieval, or transfer.
BR-15	EP-06	The user shall be able to conduct storage discovery, reservation, and item-location requests through a conversational agent.
BR-16	EP-07	The system shall be able to accept authorized programmatic storage requests from other systems when the planned integration capability is introduced.
BR-17	EP-08	The system shall be able to limit freezer configuration changes and sensitive storage access to authorized personnel.
BR-18	EP-01, EP-08	The system shall be able to onboard a new freezer with a configurable internal structure before making its storage space available for use.
BR-19	EP-09	The system shall be able to represent a Project Team associated with a project, its members, and a designated Project Lead who leads that team.
BR-20	EP-04, EP-09	The user shall be able to reserve, check in, and check out an item associated with a Project Team of which they are a member, including an item originally added by another team member, subject to storage and access restrictions.
BR-21	EP-04, EP-09	The user shall be able to view who checked in and checked out items associated with their Project Team.
BR-22	EP-01, EP-09	The user shall be able to sign in and view stored items associated with their Project Teams directly on a centralized dashboard without using the agent.
BR-23	EP-06	The user shall be able to start a new storage request from the dashboard and continue that request through the conversational agent.

6.3 Functional Requirements

Functional requirements define the observable behavior of the agent and backend together. The agent is the user facing interaction for storage queries and decisions; backend services own suitability checks, permissions, state changes, and stored records. A scan supports a physical operation and updates the same backend records.

6.3.1 Conversational Discovery and Stored-Item Lookup

ID	Epic reference	Supports	Functional requirement
FR-01	EP-01	BR-02	The system shall be able to maintain the physical location hierarchy and each freezer's configurable internal compartments and reservable positions, without requiring every freezer to use the same compartment levels.
FR-02	EP-01, EP-08	BR-02, BR-06	The system shall be able to record usable dimensions or capacity, temperature range, permitted material types, and access restrictions for each applicable storage area.
FR-03	EP-06	BR-04, BR-15	The user shall be able to describe an item and its storage requirements to the agent in natural language.
FR-04	EP-06	BR-05	The system shall be able to ask the user clarifying questions when information needed for a suitability check is missing or ambiguous.
FR-05	EP-01, EP-06	BR-01, BR-03	The user shall be able to ask the agent for current storage availability and receive only the storage information that the user is authorized to access.
FR-06	EP-02	BR-06	The system shall be able to exclude storage that fails available-capacity, physical-fit, temperature, permitted-material, or requester-authorization checks.
FR-07	EP-02	BR-04, BR-06	The system shall be able to evaluate whether a large item fits without allocating the same usable capacity to another item or reservation.
FR-08	EP-02	BR-07	The system shall be able to rank compatible available freezers by physical proximity to the applicable request origin.
FR-09	EP-02, EP-06	BR-06, BR-07	The system shall be able to explain recommended options to the user through the agent, identifying the storage location and requirements each option satisfies.
FR-10	EP-01, EP-04, EP-06	BR-12	The user shall be able to ask the agent to find a stored item by an authorized item identifier or description.
FR-11	EP-01, EP-04, EP-06	BR-12	The system shall be able to return the item's current freezer and complete recorded storage path, including its location, building, floor, room, shelf, rack, box, and position where applicable, together with its current status and recorded check-in actor.

6.3.2 Agent-Mediated Reservations

ID	Epic reference	Supports	Functional requirement
FR-12	EP-03, EP-06	BR-08	The system shall be able to present the proposed item, storage position or area, and reservation terms through the agent before asking the user to confirm.
FR-13	EP-03, EP-06	BR-08	The system shall be able to create a reservation on the user's behalf only after the user explicitly confirms the proposed reservation.
FR-14	EP-03	BR-06, BR-09	The system shall be able to recheck the requester's authorization, item compatibility, and current availability when it attempts to commit a reservation.
FR-15	EP-03	BR-09	The system shall be able to reject occupied or overlapping capacity and resolve simultaneous conflicting reservation attempts so that at most one succeeds.
FR-16	EP-03, EP-06	BR-08, BR-09	The system shall be able to tell the user whether the reservation succeeded, and provide its storage location and expiry or explain a failure and offer a new suitable option.
FR-17	EP-03	BR-10	The system shall be able to cancel a reservation with no associated check-in after two days and release its held capacity.
FR-18	EP-03, EP-06	BR-08, BR-10	The user shall be able to ask the agent for the location and current status of the user's reservations.

6.3.3 Physical Operations and Assistance

ID	Epic reference	Supports	Functional requirement
FR-19	EP-04	BR-11	The system shall be able to accept the scan associated with physical placement, verify the active reservation, and record the item as checked in at that location.
FR-20	EP-04	BR-11	The system shall be able to accept the scan associated with physical removal, record the item as checked out, and release the former occupied capacity.
FR-21	EP-04, EP-06	BR-13	The system shall be able to support relocation as check-out from the original location, conversational selection and confirmation of a new reservation, and scan-assisted check-in at the destination.
FR-22	EP-04, EP-09	BR-11, BR-12, BR-13, BR-21	The system shall be able to record the item, associated Project Team, acting person, action, storage location, and time for reservation, check-in, check-out, relocation, and expiration events.
FR-23	EP-05, EP-06	BR-14	The user shall be able to request assistance through the agent for physical storage, retrieval, or transfer and provide the details needed to perform the task.

ID	Epic reference	Supports	Functional requirement
FR-24	EP-05	BR-14	The system shall be able to make an assistance request visible or assignable to authorized personnel and record its completion and associated physical scan activity.

6.3.4 Access Administration and External Requests

ID	Epic reference	Supports	Functional requirement
FR-25	EP-08	BR-17	The system shall be able to authenticate the requester and enforce Staff and Administrator permissions for agent-initiated, scan-initiated, and backend actions.
FR-26	EP-08	BR-17	The system shall be able to withhold restricted storage details and prevent restricted reservations unless the requester has the required authorization.
FR-27	EP-08, EP-06	BR-17	The user shall be able to request freezer and restricted-area configuration changes through the agent only when assigned Administrator privileges.
FR-28	EP-06	BR-06, BR-08, BR-12	The system shall be able to base agent responses on current authorized backend results and report a reservation as completed only after the backend confirms its creation.
FR-29	EP-07	BR-16	The system shall be able to apply the same authorization, compatibility, and reservation rules to accepted programmatic requests from external systems when that capability is introduced.
FR-30	EP-08	BR-18	The system shall be able to let an authenticated Administrator initiate a new-freezer onboarding request through the agent and create a draft linked to its physical location.
FR-31	EP-01, EP-08	BR-18	The system shall be able to configure a freezer's internal storage structure from a reusable template or a custom arrangement, with only the compartment levels that apply to that freezer.
FR-32	EP-08	BR-18	The system shall be able to record capacity or dimensions, temperature characteristics, permitted material types, and access restrictions for the freezer and its applicable storage areas.
FR-33	EP-08	BR-18	The system shall be able to validate the draft configuration, present it for Administrator review, and activate its storage positions only after the Administrator confirms the configuration.
FR-34	EP-08	BR-18	The system shall be able to prevent a freezer or storage position with active reservations or stored items from being removed or changed in a way that invalidates those records.

ID	Epic reference	Supports	Functional requirement
FR-35	EP-09	BR-19	The system shall be able to maintain each Project Team's associated project, active members, and designated Project Lead, with the Project Lead recorded as a member of that team.
FR-36	EP-04, EP-09	BR-20	The system shall be able to associate each project item and its reservation with the relevant Project Team.
FR-37	EP-04, EP-09	BR-20	The user shall be able to reserve, check in, or check out a Project Team item when they are an active member of that team, regardless of which team member created the item or reservation.
FR-38	EP-04, EP-09	BR-20, BR-21	The system shall be able to check current Project Team membership and applicable freezer restrictions before allowing an item lookup, check-in, check-out, or activity-history request.
FR-39	EP-04, EP-09	BR-21	The user shall be able to view the check-in and check-out history of items associated with their Project Team, including the acting person, action, time, and recorded storage location.
FR-40	EP-01, EP-09	BR-22	The user shall be able to view items associated with their active Project Teams and their recorded storage paths on a signed-in dashboard without invoking the agent.
FR-41	EP-06	BR-23	The user shall be able to open a new-item conversation with the agent from the dashboard.

6.4 Nonfunctional Requirements

These requirements state cross-cutting quality expectations for the conversational agent, scan workflow, and backend services. Numeric performance, uptime, recovery, retention, and accessibility targets remain matters for stakeholder agreement in the BRD appendix.

ID	Epic reference	Quality area	Nonfunctional requirement
NFR-01	EP-03, EP-04	Reliability	The system shall be able to preserve a consistent reservation, occupancy, and item-location state when a request, scan, or backend transaction fails or is interrupted.
NFR-02	EP-06, EP-07, EP-08	Security	The system shall be able to enforce access restrictions consistently across conversations, scanning, administration, and authorized external requests.
NFR-03	EP-03, EP-04, EP-08	Auditability	The system shall be able to retain attributable records of reservation confirmations, material activity, and administrator configuration changes.
NFR-04	EP-02, EP-03, EP-06	Usability	The system shall be able to communicate missing information, suitability, availability, reservation outcome, exact storage location, and the next physical action clearly in conversation.

7. Business Rules

ID	Business rule
RULE-01	An item must have a reservation before physical check-in.
RULE-02	Active reservations and occupancy cannot overlap incompatible uses of the same capacity.
RULE-03	An unused reservation is cancelled after two days and the released capacity becomes available.
RULE-04	Only an Administrator can change a freezer structure or its restrictions.
RULE-05	Restricted storage can be used only by authorized personnel.
RULE-06	A location must meet physical and environmental conditions before proximity is used to rank it.
RULE-07	Relocation consists of checkout, a new reservation, and check-in.
RULE-08	Conversational recommendations use the same current storage data and business rules as the standard interface.

8. Assumptions, Dependencies, and Constraints

8.1 Planning Assumptions

Assumption	Effect if inaccurate
Users provide accurate item information and requirements.	Recommendations can be unsuitable.
Administrators keep the digital freezer configuration aligned with the physical layout.	Displayed availability can be wrong.
Staff scan during physical check-in and check-out.	The digital record can diverge from reality.
Physical location information can be obtained for proximity comparison.	The closest-freezer ranking can be misleading.

8.2 Dependencies

The solution depends on accurate freezer inventory, an identity and authorization mechanism, scan-capable workflows at the laboratory, reliable remote application access, and agreed location information for calculating proximity.

8.3 Constraints

ID	Constraint
C-01	Only administrators can onboard, remove, or change freezers and their structures.
C-02	Sensitive storage is restricted according to assigned authorization.
C-03	An active reservation holds its capacity until check-in, cancellation, or expiration after two days.
C-04	Freezer layouts and usable space are heterogeneous.
C-05	Empty capacity alone does not establish item compatibility.

9. Use Cases

Symbol	Meaning
Stick figure	An external person or system that interacts with this system.
Oval	A goal or capability the system provides.
Rectangle	The system boundary. Its use cases are inside; actors are outside.
Solid line	The actor participates in that use case. It does not show step order.
Dashed arrow	A relationship between use cases; the arrow points to its target.
«include»	The source use case always uses the target use case.
«extend»	The source adds behavior to the target when a condition applies.

Figure 6 Legend for the use case diagrams

9.1 Use Case 1: Finding and reserving storage

Staff describe their material to the agent. The system can ask for missing details, evaluate backend storage records, recommend suitable space, and obtain explicit confirmation before attempting the reservation.

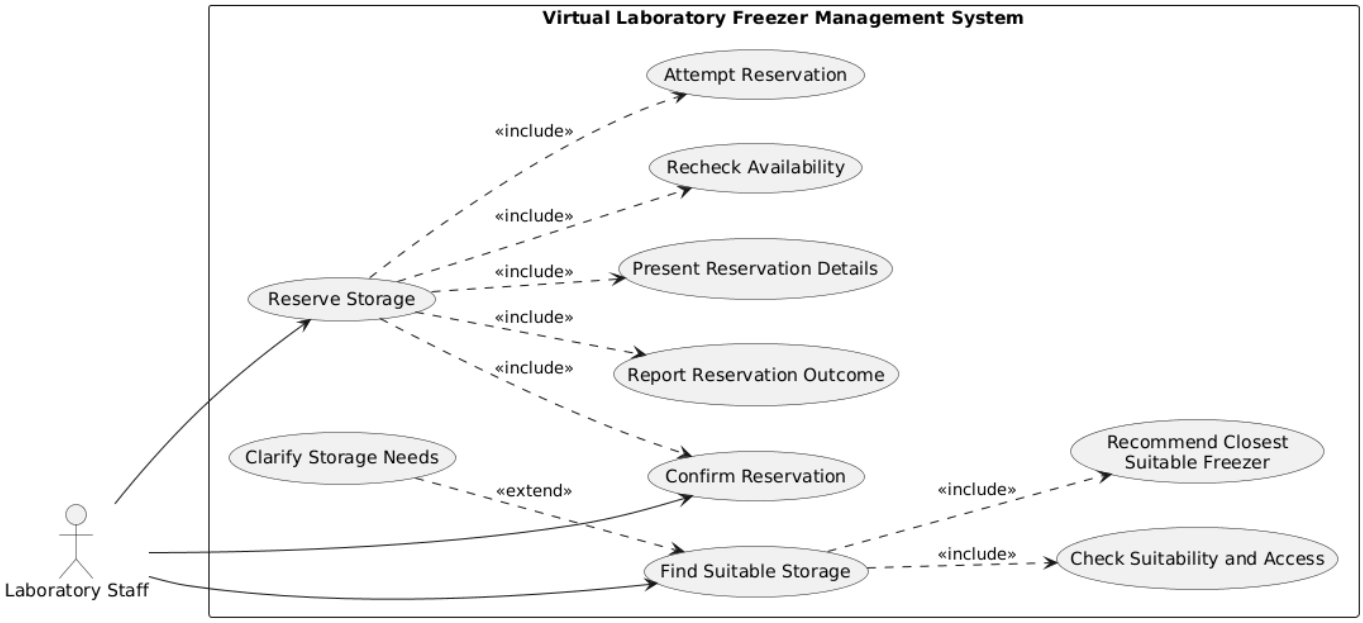


Figure 7 Use case diagram for Finding and reserving storage

9.2 Use Case 2: Storing, locating, and removing an item

Description: Staff can ask the agent for an item's current freezer and complete storage path. Physical placement and removal use scans. Moving an item uses check-out at the original location, a confirmed destination reservation, and check-in at the new location.

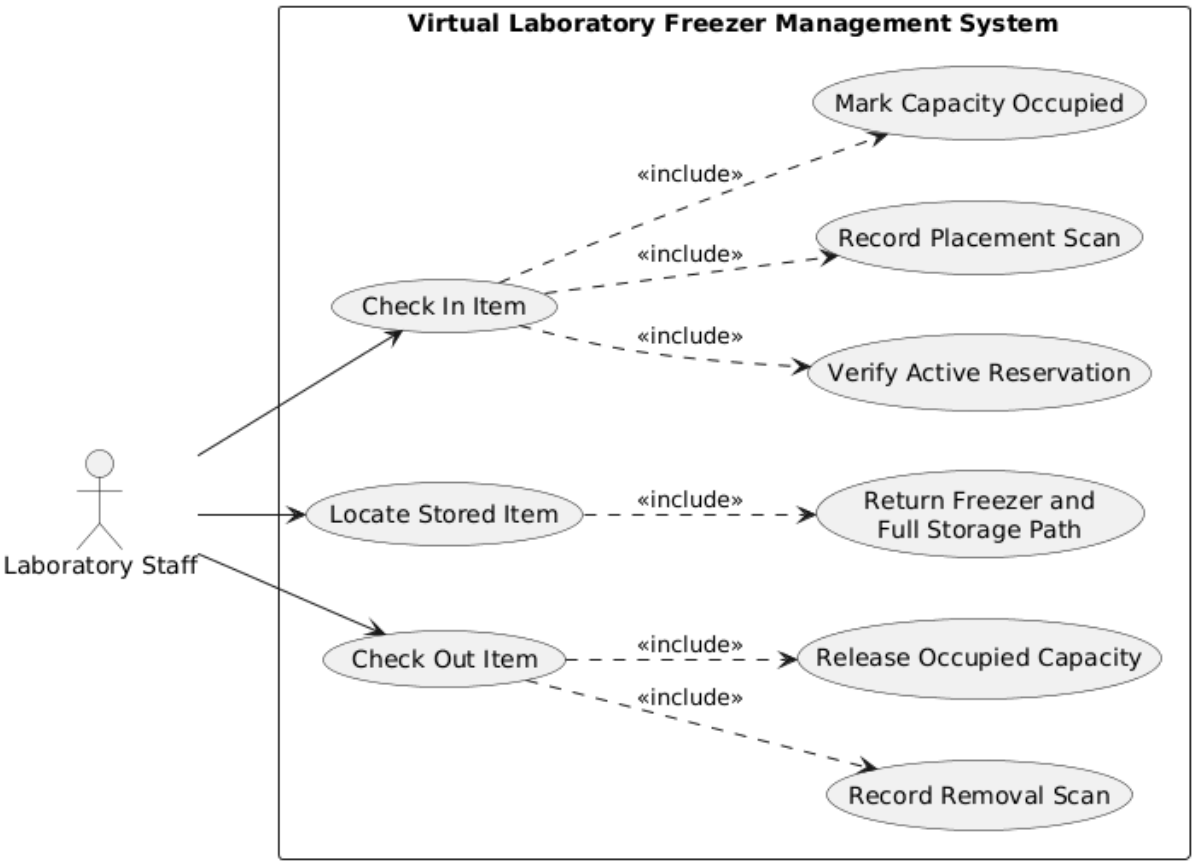


Figure 8 Use Case diagram for Storing, locating and removing an item

9.3 Use Case 3: Requesting assistance

Description: Staff ask the agent for help with storage, retrieval, or transfer. Authorized personnel can view the request, perform the physical work, and record the relevant scan and completion.

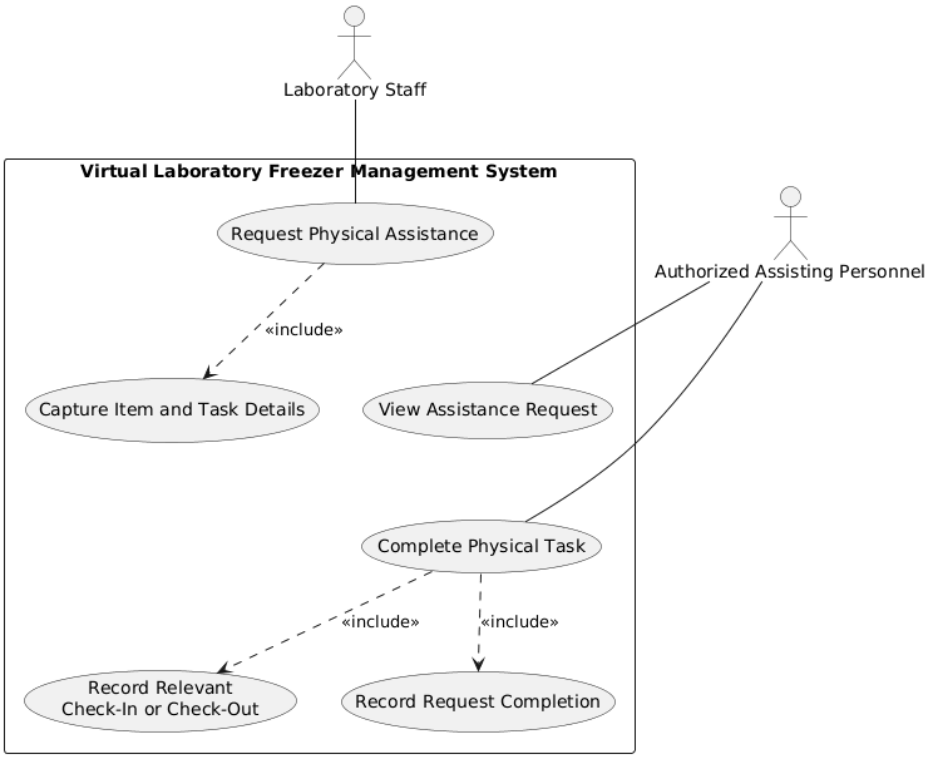


Figure 9 Use Case Diagram for Requesting Assistance

9.4 Use Case 4: Administering freezers and restricted storage

Description: An administrator requests configuration changes through the agent. The system checks administrator permissions before changing freezer structures, storage attributes, or restrictions.

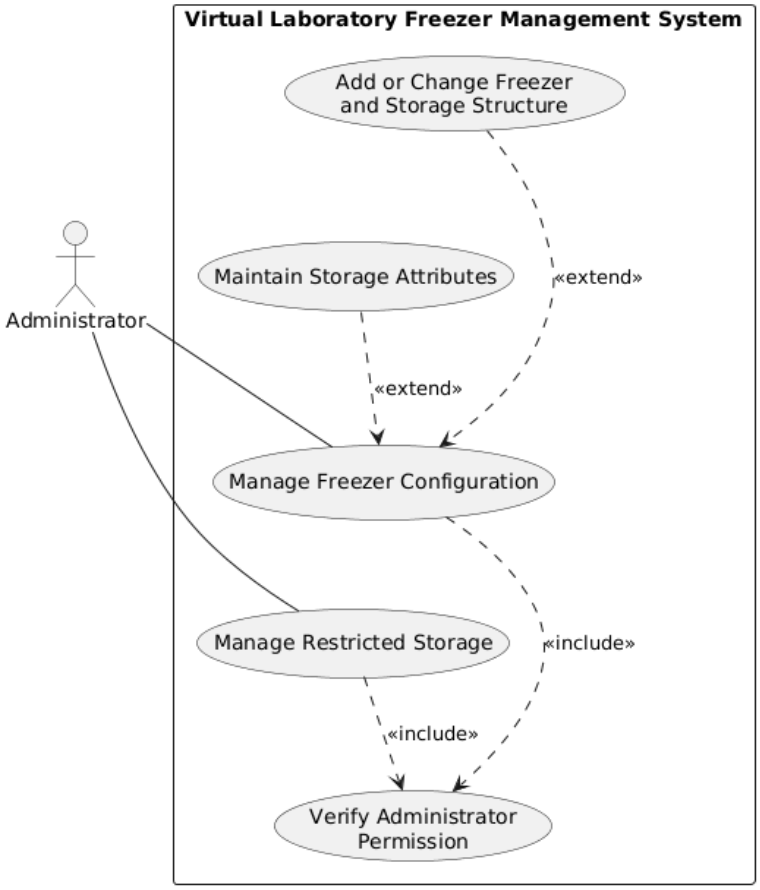


Figure 10 Use Case diagram for Administering freezers and restricted storage

10. High level system design diagram

The agent handles conversation and confirmation; backend services check permissions and perform the work. The single records symbol represents freezer structures, item locations, reservations, occupancy, assistance, and activity history. The dashed path marks future external integration.

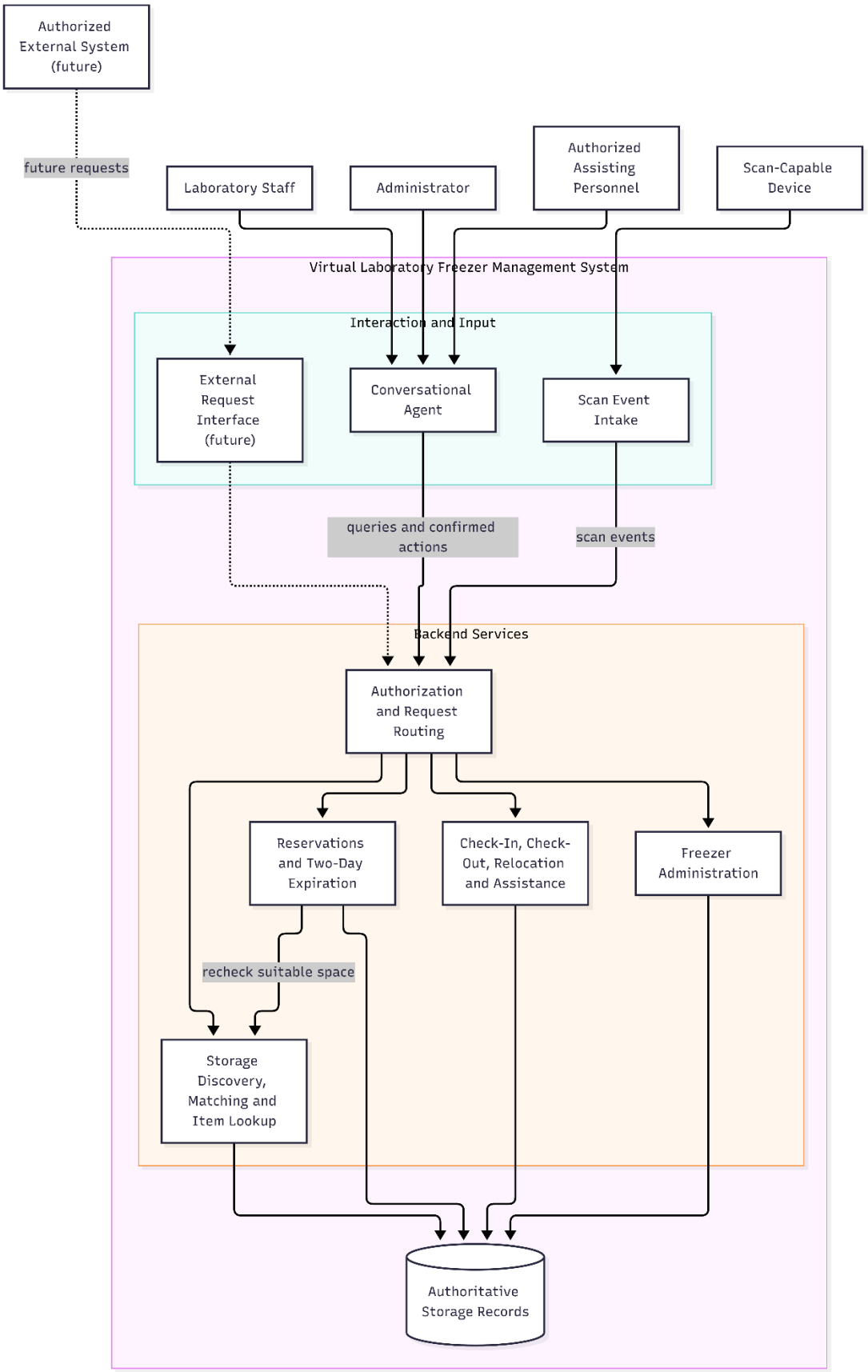


Figure 11 High level system design diagram

11. Business Risks

Risk	Description	Likelihood	Impact	Mitigation / Response	Status
R-01	Digital storage state diverges from the physical freezer because users skip or incorrectly perform scanning/check-in/check-out.	Medium	High	Require scan-assisted workflow; provide reconciliation/admin correction capability.	Proposed
R-02	Two users attempt to reserve the same capacity at nearly the same time.	Medium	High	Use conflict-safe reservation logic and verify final state atomically.	Confirmed risk
R-03	Incorrect item requirements lead to unsuitable storage recommendations.	Medium	High	Validate required fields and present compatibility criteria before reservation.	Proposed
R-04	Incorrect freezer configuration or restrictions lead to bad recommendations.	Medium	High	Restrict configuration to admins; provide validation/review of configuration data.	Proposed
R-05	Agent recommends space not supported by authoritative storage data.	Medium	High	Constrain agent actions to backend data and business rules; require authorization checks.	Proposed
R-06	Physical proximity data is incomplete or inaccurate, so “closest” recommendations are misleading.	Medium	Medium	Define location model and distance method; expose location context to user.	Proposed
R-07	Reserved capacity remains unused until expiration, reducing short-term availability.	Medium	Medium	Automatic two-day expiration; consider reminders/manual cancellation during refinement.	Partly confirmed
R-08	Unauthorized users gain visibility or access to sensitive storage.	Low/Medium	High	Role-based authorization and restricted-storage controls.	Confirmed concern
R-09	Scanning technology is unavailable or fails during physical operations.	Medium	Medium	Define supported devices and a controlled fallback process during design.	Proposed
R-10	Scope expands rapidly around AI agent integrations before core storage workflows are stable.	Medium	Medium	Prioritize core storage, reservation, and check-in/out requirements; phase integrations.	Proposed

12. Success and Acceptance Criteria

12.1 Business Success Criteria

- Authorized staff can identify suitable available storage without first visiting each physical freezer.
- The system provides centralized visibility across supported freezer locations and structures.
- Reservations prevent conflicting use of the same capacity and unused reservations release after two days.
- Users can check material in and out using the defined scan-assisted process.
- Large items are recommended only where sufficient compatible capacity exists.
- Restricted storage is not offered to users who are not authorized to use it.
- Administrators can centrally manage freezer structures and storage restrictions.
- The conversational agent can interpret a storage request and return options based on authoritative system data.

12.2 High-Level Business Acceptance Scenarios

ID	Scenario	Expected Business Outcome
AC-01	A staff member provides storage needs for an item.	The system returns only available locations that satisfy applicable capacity, temperature, storage-type, and authorization rules.
AC-02	Two users submit a reservation request for the same position concurrently.	At most one reservation succeeds; the other user receives an unavailable/conflict outcome and may select another option.
AC-03	A reservation remains unused for two days.	The reservation is cancelled and the capacity becomes available to other users.
AC-04	A user requests storage for a large item.	The system does not recommend a location that lacks sufficient capacity.
AC-05	A non-authorized user searches for storage that includes an admin-restricted freezer.	The restricted storage is not offered as a usable option to that user.
AC-06	A user physically places an item after reservation and completes the scan/check-in flow.	The selected location becomes occupied and the stored item is associated with the location and check-in user.
AC-07	A user removes an item and completes the scan/check-out flow.	The item is no longer checked in at that location and the released capacity becomes available as appropriate.
AC-08	A user needs to move an item.	The workflow supports checkout from the original location, reservation of a suitable new location, and check-in at the destination.
AC-09	A user asks the conversational agent for storage in natural language.	The agent identifies requirements, queries authoritative storage information, and returns suitable options rather than inventing availability.
AC-10	A user cannot perform a needed physical storage or transfer action.	The system allows an assistance request to be raised; detailed routing/assignment behavior is subject to further definition.
AC-11	The agent recommends suitable space, but the user has not explicitly confirmed a reservation.	No reservation is created. After the user confirms the proposed item and location, the system attempts the reservation and reports the backend result.

ID	Scenario	Expected Business Outcome
AC-12	An authorized user asks where an item is stored.	The system returns the item's current status, freezer, and complete recorded storage path, including its position where applicable.
AC-13	Suitable capacity exists in multiple freezers.	The system recommends the physically closest suitable available freezer based on the agreed request origin, after applying compatibility and access rules.
AC-14	An Administrator changes a freezer configuration or storage restriction.	The system applies the authorized change and uses the updated configuration in subsequent storage searches and reservations.
AC-15	A user gives the agent an incomplete storage request.	The agent asks for the missing details needed to evaluate suitability before recommending or reserving space.
AC-16	An authorized external system submits a storage request when the future integration capability is introduced.	The backend applies the same authorization, compatibility, availability, and reservation rules used for agent-initiated requests.
AC-17	An Administrator onboards a freezer with a layout different from existing freezers.	The system records its physical location, storage structure, capacity, temperature characteristics, and restrictions; its usable space then appears in suitable searches.
AC-18	An Administrator onboards a freezer with shelves, racks, boxes, or positions that differ in size or permitted use.	The system represents those differences and recommends or reserves only positions that meet an item's requirements.
AC-19	A Project Lead creates or manages a project team and its membership.	The system associates the team with its Project Lead and applies current membership when authorizing access to the team's items and activity records.
AC-20	A project team member checks in or checks out an item originally added by another member of the same team.	The permitted action succeeds, storage and capacity records are updated, and the person who performed the action is recorded.
AC-21	A project team member requests the check-in and check-out history for a team item.	The system shows that team member the attributable activity log, including who performed each action and when, while preventing access by unauthorized users.

13. Requirements Traceability Matrix

The matrix links each business requirement to its originating stakeholder topic, implementing functional requirements, and high-level acceptance scenarios. “Appendix A” refers to the stakeholder interview summary.

Business requirement	Epic(s)	Source in Appendix A	Functional requirement(s)	Acceptance scenario(s)
BR-01	EP-01, EP-06	Availability; Conversational agent	FR-05	AC-09
BR-02	EP-01	Physical hierarchy; Freezer variability	FR-01, FR-02	AC-01, AC-12
BR-03	EP-01, EP-06	Availability	FR-05	AC-01, AC-03
BR-04	EP-02	Freezer variability; Material types	FR-03, FR-07	AC-04
BR-05	EP-02, EP-06	Conversational agent	FR-04	AC-15
BR-06	EP-02	Compatibility	FR-02, FR-06, FR-07, FR-09, FR-14, FR-28	AC-01, AC-04, AC-05
BR-07	EP-02	Closest freezer	FR-08, FR-09	AC-13
BR-08	EP-03, EP-06	Reservation; Conversational agent	FR-12, FR-13, FR-16, FR-18, FR-28	AC-11
BR-09	EP-03	Concurrency	FR-14, FR-15, FR-16	AC-02
BR-10	EP-03	Reservation	FR-17, FR-18	AC-03
BR-11	EP-04	Scanning	FR-19, FR-20, FR-22	AC-06, AC-07
BR-12	EP-01, EP-04, EP-06	Search / traceability; Conversational agent	FR-10, FR-11, FR-22, FR-28	AC-12
BR-13	EP-04, EP-06	Movement	FR-21, FR-22	AC-08
BR-14	EP-05, EP-06	Movement; assistance request	FR-23, FR-24	AC-10
BR-15	EP-06	Conversational agent	FR-03, FR-04, FR-10, FR-12, FR-13, FR-28	AC-09, AC-11, AC-12, AC-15
BR-16	EP-07	External requests	FR-29	AC-16
BR-17	EP-08	Administrator; Staff access	FR-25, FR-26, FR-27	AC-05, AC-14

Appendix A. Stakeholder Interview Summary

Topic	Interview Finding
Primary users	Staff, Project Lead, Administrator.
Staff access	Staff use the system to add/remove items from storage and perform storage workflows.
Project Lead	Leads development/oversight of the system; exact operational permissions are not yet fully defined.
Administrator	Only administrators can add/remove/modify freezers/configuration; some sensitive storage is admin-only.
Physical hierarchy	Location → Building → Floor → Room → Freezer → Shelf → Rack → Box.
Freezer variability	Freezers can differ in type/layout and may reserve space for larger objects.
Availability	Availability can exist at freezer, shelf, rack, box, or individual slot/position level.
Material types	No exhaustive list; the lab may store many kinds of items.
Reservation	Storage must be reserved before placement. Unused reservations are cancelled after two days and the space is released.
Closest freezer	Recommend the literally closest suitable freezer; this may be on the same floor/building or at a different building/location.
Concurrency	The system must not allow multiple users to reserve the same location and must handle race conditions cleanly.
Compatibility	Freezers can have different temperatures and item restrictions; suitability includes capacity for larger items.
Scanning	Use an app/scanning process when physically adding or removing material.
Movement	Move by checking out, reserving a new location, and checking in; users can raise a help request when they cannot perform the physical action.
Search / traceability	The system must capture who checked in and what was checked in; users provide information before reserving.
Conversational agent	Users should converse with an agent, describe storage requirements, and have the agent check backend/database availability.
External requests	The agent/service should be able to support requests from other systems; Claude/MCP was mentioned only as a possible technical direction.

Appendix B. Technology Notes for Later SDLC Phases

The following notes preserve technology related ideas raised during requirements discussions without treating them as approved business requirements:

- A conversational/agentic interface is a confirmed business capability; the specific AI model/vendor is not selected by this BRD.
- Claude and MCP were mentioned as possible approaches for external-system interaction. They should be evaluated during architecture/design against security, interoperability, maintainability, cost, and organizational constraints.
- The scanning mechanism is not yet selected. QR, barcode, RFID, NFC, or another approach should not be assumed until the scanning workflow and hardware environment are confirmed.
- Database, API, cloud, frontend, and deployment choices belong in architecture and detailed design documents after this BRD is validated.

Appendix C. Glossary and Acronyms

Term	Definition
BRD	Business Requirements Document.
Storage Location	A reservable or viewable physical area within the freezer hierarchy, such as a freezer, shelf, rack, box, or position/slot as applicable.
Available	Capacity that is not occupied, not actively reserved, and permitted for the requesting user/material.
Reserved	Capacity temporarily held for a user before physical check in.
Occupied	Capacity associated with checked-in material.
Check-In	Business action that records physical placement of material into its reserved storage location.
Check-Out	Business action that records physical removal of material from its storage location.
Restricted Storage	Storage whose use is limited to administrators or other explicitly authorized users.
Conversational Agent	An interface that accepts natural-language storage requests and uses system data/business rules to help identify suitable storage.
MCP	Model Context Protocol; mentioned as a possible future technical integration approach, not mandated by this BRD.
RBAC	Role-Based Access Control.

Appendix D. Proposed Technology Stack

These are implementation proposals, not mandatory business requirements. The frontend is a conversational interface; users do not browse freezers or select storage through a separate click based workflow.

Layer	Existing proposed choice	Alternative choice for Axle	Reason for the alternative
Frontend	React, TypeScript, Vite	Keep React, TypeScript, Vite; host the built UI on Azure Static Web Apps	The UI does not need to change with the cloud provider. Azure Static Web Apps supports React deployment.
Backend and APIs	Python, Django, Django REST Framework	Python/FastAPI, or Node.js/TypeScript with NestJS	Both fit the team's stated experience. Select **one** as the authoritative core API so reservation and permission rules are not duplicated across two backends. FastAPI supports reusable security dependencies; NestJS provides a structured Node.js backend.
Conversational agent	Claude through Amazon Bedrock; tools call Django	Azure OpenAI in Microsoft Foundry; tools call the selected backend	The model handles language and proposes tool calls; application code executes them and enforces confirmation and authorization. Model access and organizational approval still need validation.
Storage matching	Python service within Django	Matching module within FastAPI or NestJS	Keep compatibility checks and proximity ranking in the same authoritative backend chosen above.
Database	PostgreSQL on Amazon RDS	Azure Database for PostgreSQL Flexible Server	Retains PostgreSQL and its transaction approach while moving the managed database to Azure.
Reservation expiration	EventBridge Scheduler; idempotent Django task	Azure Functions timer trigger; idempotent expiration task	A scheduled task checks reservations whose stored expiration time has passed and releases only unused holds. The timer interval determines how soon after the two-day point a hold is processed.
Physical scanning	QR labels; @zxing/browser	Keep QR labels and @zxing/browser	Browser scanning is independent of AWS or Azure. QR remains a proposed mechanism until the laboratory scanning decision is confirmed.
Identity and access	Organizational OIDC SSO; Django roles	Microsoft Entra ID using OIDC; roles and Project Team membership enforced in the selected backend	Entra can provide organizational sign-in; the application still decides which project items, logs, and freezer areas each person may access.

Deployment	Docker on ECS/Fargate	Docker on Azure Container Apps; React build on Azure Static Web Apps	Container Apps can host the API without managing server instances and can run scheduled jobs if that becomes preferable to a timer function.
Delivery	Terraform; GitHub Actions	Terraform with the Azure provider; GitHub Actions or Axle's existing Azure DevOps pipeline	The infrastructure-as-code approach remains useful; the deployment pipeline should follow the organization's established repository and approval process.
Monitoring and audit	CloudWatch Logs; activity records in PostgreSQL	Azure Monitor/Application Insights; attributable activity records in Azure PostgreSQL	Application telemetry helps operators diagnose failures. The project-team history should come from durable business activity records with team-based access checks.